

Calibre: Probability calibration and evaluation in Python

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Abstract

Calibre combines probability calibration methods with tools for evaluating their accuracy, calibration, and effect on score ordering. Its common interface covers isotonic and centered isotonic regression, monotone splines, and methods that relax monotonicity. Evaluation includes proper scores, isotonic reliability diagrams and score decompositions, calibration-error estimators, and pointwise uncertainty bands. A reproducible comparison uses 7 synthetic designs and one real dataset across 30 seeds. In the overconfidence design, centered isotonic regression retains an average of 1514 distinct test predictions compared with 49 for isotonic regression, with mean Brier scores of 0.1527 and 0.1530, respectively. Temperature scaling estimates the known probabilities more accurately in that design. These results illustrate why prediction granularity, calibration error, and predictive accuracy should be reported separately. Calibre’s contribution is an implemented and tested workflow for making those comparisons; its methods do not guarantee better predictions or decision utility in every application.

1 Calibration as a comparison problem

Calibre lets researchers compare calibration maps using the same fitting interface and a common set of evaluation tools. A researcher can fit an isotonic map, replace it with centered isotonic regression or a monotone spline, and inspect changes in proper scores, reliability, and the distinctions between predicted probabilities. The software joins these operations without reducing their outputs to a single calibration verdict.

For a binary outcome Y , a forecast P is calibrated when $\mathbb{E}[Y | P] = P$. A constant forecast at the population event rate can satisfy that condition while distinguishing no observations. Proper scoring rules evaluate both calibration and the information in a forecast (DeGroot and Fienberg, 1983; Gneiting and Raftery, 2007). Calibration error alone therefore cannot determine whether recalibration has improved predictions.

Isotonic regression illustrates the distinction. Pooling adjacent scores can remove noise but also assign the same forecast to observations with different input scores. Interpolating between pooled estimates can retain more distinctions, but those distinctions may not correspond to differences in event probabilities. Whether they help is an empirical question for observations not used to fit the map. Calibre reports distinct-value counts alongside proper scores and ranking measures so that retaining more values is not mistaken for improving predictions.

The contribution is software integration and validation. Centered isotonic regression, nearly isotonic regression, monotone spline bases, and isotonic reliability diagrams have established methodological sources (Oron and Flournoy, 2017; Tibshirani et al., 2011; Ramsay, 1988; Dimitriadis et al., 2021). Calibre supplies Python implementations or wrappers, shared numerical primitives, consistent input validation, and executable comparisons. It complements scikit-learn’s calibration interface with these methods and evaluation tools (scikit-learn developers, 2026). The benchmark below evaluates the implemented defaults, including their internal tuning, rather than claiming that one family dominates after an exhaustive search.

2 Fitting calibration maps

Let (s_i, y_i, w_i) denote a score, binary outcome, and nonnegative observation weight. Zero-weight observations do not contribute to fitting. Tied input scores are pooled before interpolation: for unique score x_j , the target $\bar{y}_j$ is the weighted outcome mean and W_j is total weight. A deterministic map cannot separate observations that have exactly the same input score. The methods in Table 1 differ in their treatment of distinct scores, their shape restrictions, and their tuning objectives.

Table 1: Implemented binary calibration methods. “Linear” describes interpolation between fitted anchors, with constant extrapolation beyond the fitted support. Non-decreasing maps can contain ties. Neither nearly isotonic nor a negative increment bound guarantees monotonicity. CDI-ISO is experimental and described in Appendix B.

Method	Fit	Shape and prediction	Tuning
Isotonic	Weighted least squares under ordering	Non-decreasing; linear	None
Centered isotonic	Pooled rates at block anchors	Non-decreasing; linear	None
Nearly isotonic	Squared loss plus penalty on decreases	Decreases permitted; linear	Penalty selected by CV by default
I-spline	Penalized Bernoulli or squared loss	Non-decreasing spline	Knot count and penalty selected by CV by default
Relaxed PAVA	Squared loss with adjacent increment bound	Depends on bound; linear	User must specify bound
CDI-ISO	Squared loss with local increment bounds	Decreases possible; step	User supplies thresholds

2.1 Isotonic and centered isotonic regression

Weighted isotonic regression minimizes

$$\sum_{j=1}^m W_j (\bar{y}_j - z_j)^2 \quad \text{subject to} \quad z_1 \leq \cdots \leq z_m. \quad (1)$$

The pool-adjacent-violators algorithm (PAVA) solves this ordered problem in linear time (Ayer et al., 1955). Calibre’s isotonic estimator wraps scikit-learn. Fitted values contain constant blocks; prediction interpolates linearly between fitted score knots.

Centered isotonic regression collapses a pooled block to a score anchor and interpolates between successive anchors (Oron and Flournoy, 2017). Interior anchors are weighted score centroids. The implementation preserves flat boundary blocks at fitted probabilities zero and one, and holds endpoint values constant outside the anchor range. Centering is not a minimizer of Equation 1. It imposes an interpolation assumption between pooled estimates, which can preserve distinctions without establishing that the underlying event probability varies within every block.

2.2 Monotone splines and relaxed ordering

The spline estimator fits

$$\ell\{g(s)\} = \theta + \sum_{k=1}^K \delta_k I_k(s), \quad \delta_k \geq 0,$$

where I_k are non-decreasing I-spline basis functions and ℓ is the link. The default logit link uses penalized Bernoulli loss; the identity link uses penalized squared loss. A penalty on successive coefficient differences controls roughness. The basis and penalty follow established spline constructions (Ramsay, 1988; Eilers and Marx, 1996). Nonnegative coefficients guarantee a non-decreasing map, not a strictly increasing one. Cross-validation selects knot count and penalty before refitting on all calibration observations. The default scoring rule is log loss for the logit link and squared loss for the identity link.

Nearly isotonic regression replaces the ordering constraint with a penalty:

$$\frac{1}{2} \sum_j W_j (\bar{y}_j - z_j)^2 + \lambda \sum_{j=1}^{m-1} (z_j - z_{j+1})_+.$$

Calibre implements the modified-PAVA solution path on the original penalty scale (Tibshirani et al., 2011). By default, cross-validation selects λ ; the search holds the penalty per unit observation weight constant between training folds and the full fit. Finite penalties can reverse ranks. A small penalty can also fit outcome noise, so the existence of a nonmonotone population curve does not establish that the default estimator will recover it.

Relaxed PAVA instead imposes $z_{j+1} - z_j \geq L$, with a required, user-specified `min_increment`. Zero recovers isotonic regression, negative values permit bounded decreases, and positive values require separation before clipping. The bound is per adjacent unique score, not per unit change in score. Appendix A gives the reduction and its limitations.

3 Evaluation and use

3.1 Proper scores, reliability, and granularity

Calibre reports Brier score, $n^{-1} \sum_i (y_i - p_i)^2$, and supports logarithmic score in selection and decomposition. Lower scores are better. Its CORP reliability diagram fits isotonic regression of observed outcomes on forecast probabilities (Dimitriadis et al., 2021). This evaluation fit is a diagnostic of the forecasts on the evaluation sample, not a new deployed calibrator. The CORP decomposition is

$$\bar{S} = \text{MCB} - \text{DSC} + \text{UNC},$$

where MCB is the score reduction from isotonic recalibration on that sample, DSC is the improvement of those recalibrated forecasts over the sample event-rate forecast, and UNC is the latter forecast’s score. These finite-sample components are not unbiased population quantities. Their identity is useful for understanding a score difference, but a small estimated MCB alone does not establish good predictive performance.

Granularity is the number or fraction of distinct forecast values. It is different from CORP discrimination and from AUC. A strictly increasing transformation preserves rankings and AUC; a non-decreasing transformation can introduce ties, and a nearly isotonic transformation can reverse rankings. The benchmark counts distinct values after rounding probabilities to six decimal places. The one-call report counts unique predictions under its own documented convention; users should state the convention when comparing counts.

3.2 Calibration-error estimators and uncertainty

Binned error depends on both the norm and the binning rule. Calibre’s expected calibration error is an ℓ_1 gap over uniform-width bins. Its plugin estimator allows a matched ℓ_2 comparison on approximately equal-mass bins without splitting tied predictions. The debiased estimator subtracts a binomial variance term on the squared scale (Kumar et al., 2019). Taking a square root after flooring at zero reintroduces bias near zero; the correction does not remove binning approximation error. The benchmark records this default nonnegative root, so it should not be described as an unbiased error estimate.

The monotonic sweep chooses a bin count using monotonicity of empirical bin event rates (Roelofs et al., 2022). Its interpretation depends on a non-decreasing population calibration curve, an assumption deliberately violated by one benchmark design. Smooth calibration error uses kernel smoothing and an automatic bandwidth rule following the published reference implementation (Błasiok and Nakkiran, 2024). These measures answer different questions and should not be ranked by their numerical size as though they shared one scale and construction.

The evaluation API provides pointwise consistency bands under the calibration null and pointwise confidence bands around a fitted CORP curve. The former simulate outcomes from the supplied forecast probabilities; the latter simulate from PAV-recalibrated probabilities and include a boundary correction. Both follow the resampling procedures associated with CORP (Dimitriadis et al., 2021). Neither is a simultaneous envelope or a global calibration test. A separate bootstrap function estimates intervals for scalar metrics; such intervals condition on the supplied predictions and do not automatically include uncertainty from training the original model.

Plateau diagnostics describe score spans, calibrated values, and observation counts. Their support labels use conventional count thresholds. They do not test whether the population curve is flat, measure within-plateau discrimination, or justify breaking a tie.

3.3 A separated fitting and evaluation workflow

Calibrators expose `fit(X, y, sample_weight=None)` and `transform(X)`. Base-model scores used to fit a calibrator should come from observations excluded from that base model’s training, and final evaluation should use a further held-out sample. The following executable example constructs scores directly, fits on one independent subset, and evaluates on another:

```
import numpy as np

from calibre import CenteredIsotonicCalibrator, calibration_report

rng = np.random.default_rng(42)
latent = rng.normal(size=2000)
probability = 1 / (1 + np.exp(-latent))
scores = 1 / (1 + np.exp(-1.8 * latent))
y = rng.binomial(1, probability)

calibrator = CenteredIsotonicCalibrator().fit(scores[:1000], y[:1000])
predictions = calibrator.transform(scores[1000:])
report = calibration_report(y[1000:], predictions)
print(report)
```

The report assembles metrics and score components without fitting another deployment map or returning a hypothesis-test verdict. Plotting functions consume computed results or fitted calibrators, with matplotlib installed through the optional plotting extra. Cross-validation helpers support shuffled folds for independent, exchangeable observations; they do not supply grouped or temporal splitting.

The multiclass interface includes temperature scaling, classwise reliability and score decomposition,

and top-label and classwise calibration errors. Temperature scaling applies a common positive temperature and preserves the maximizing class. Classwise calibration and top-label calibration are distinct properties; neither establishes calibration of the entire probability vector. The binary benchmark below does not establish comparative multiclass performance.

4 A reproducible comparison

4.1 Design and estimands

The offline grid crosses 7 synthetic designs with an identity score model and the Wisconsin diagnostic breast-cancer dataset with logistic regression, a random forest, and histogram gradient boosting. The real dataset is distributed by scikit-learn; its positive label denotes benign cases (Wolberg et al., 1993). There are 10 dataset/model configurations, 8 methods, and 30 seeds, producing 2400 method-level records. The Calibre isotonic wrapper is included as an implementation check against scikit-learn isotonic, not as an independent competing algorithm.

In the overconfident, underconfident, and prior-offset designs, $Z \sim N(0, 2^2)$ and the true probability is $\sigma(Z)$, where σ is the logistic function. Reported scores are respectively $\sigma(1.8Z)$, $\sigma(0.6Z)$, and $\sigma(Z + 0.8)$. The tied-score design rounds the overconfident score to two decimal places. The rare-event design draws $Z \sim N(-4.6, 1.5^2)$ and reports $\sigma(1.6Z)$ for truth $\sigma(Z)$; it does not enforce a fixed prevalence. The small-sample design uses the overconfidence construction with 300 rather than 4,000 observations. The nonmonotone design draws $s \sim U(0, 1)$ and clips $s + 0.25 \sin(2\pi s)$ to $[0.01, 0.99]$ to obtain the event probability. All other synthetic designs have 4,000 observations per seed. Outcomes are Bernoulli draws conditional on these probabilities.

The benchmark holds out 40% of each dataset. For real data, 5 stratified folds within the fitting sample generate out-of-fold base-model probabilities. Preprocessing is fitted inside each fold. Calibrators fit those scores, and the base model is refitted on the full fitting sample before scoring the test sample. Thus calibration uses scores from somewhat smaller training sets than deployment; the experiment measures this particular workflow. Synthetic designs split the constructed scores directly. Within each seed and configuration, every calibrator receives identical inputs.

Comparators are uncalibrated scores, isotonic regression, logistic recalibration of log-odds, and temperature scaling. The logistic comparator uses scikit-learn logistic regression with a near-unpenalized fit to log-odds. The temperature comparator uses a bounded scalar optimization of log temperature. Neither calls `CalibratedClassifierCV`; scikit-learn’s sigmoid wrapper instead uses the base estimator’s decision function or probability output (scikit-learn developers, 2026). Calibre’s centered, spline, and nearly isotonic estimators run at library defaults. Both spline and nearly isotonic defaults include internal cross-validation, so tuning budgets differ. Relaxed PAVA and CDI-ISO require application-specific inputs and are not in this defaults-only grid.

Brier score and CORP miscalibration are the configured primary measures. Supporting measures include logarithmic score, AUC, granularity, and runtime. Plugin and debiased binned errors use 15 requested equal-mass bins and the ℓ_2 norm; sweep error also uses ℓ_2 . Mean absolute error against the generating probability is available only for synthetic data. With rounded scores, this is error against the latent individual probability, not the conditional mean given the coarsened score; some of it cannot be removed by any deterministic score map.

Each comparison subtracts a method’s Brier score from isotonic’s within the same seed. Positive differences favor the method. Intervals are percentile intervals from 2000 bootstrap resamples of these 30 paired differences. Synthetic seeds regenerate data; real-data seeds repeatedly split the same observations. The latter intervals describe split and algorithm variability conditional on this dataset, not uncertainty over independent datasets or clinical populations. The intervals are descriptive and are not adjusted for multiple comparisons.

4.2 Accuracy and granularity need separate evidence

Centered isotonic and the spline retain more distinct forecasts than isotonic in the overconfidence design, with lower mean Brier scores (Table 2). Centered isotonic produces 1514 distinct test values on average versus 49 for isotonic. Their mean Brier scores are 0.1527 and 0.1530. This joint comparison supports a limited claim: under this design, retaining more distinctions need not worsen average held-out score. Their CORP miscalibration components are nevertheless larger than isotonic’s: the lower Brier scores do not imply an improvement on every calibration diagnostic. Additional distinctions are not always useful.

Table 2: Overconfidence design, averaged over 30 seeds. Brier and MCB use the probability-squared scale. Distinct counts use six-decimal rounding and are rounded here to integers. Truth MAE is mean absolute error against the generating probability. Lower scores and errors are better; a larger distinct count is not itself a performance gain. The duplicate isotonic wrapper is omitted.

Method	Brier	MCB	Distinct	Truth MAE
Uncalibrated	0.1604	0.0121	1594	0.0808
Isotonic	0.1530	0.0025	49	0.0255
Logistic (log-odds)	0.1521	0.0039	1599	0.0064
Temperature	0.1522	0.0039	1599	0.0040
Centered isotonic	0.1527	0.0044	1514	0.0205
I-spline	0.1524	0.0042	1595	0.0175
Nearly isotonic	0.1531	0.0025	52	0.0259

Temperature scaling has mean absolute error 0.0040 against the generating probability, compared with 0.0175 for the spline and 0.0205 for centered isotonic. Here the distortion is exactly a temperature change, so the one-parameter family contains the true inverse map. This comparison supports including simple parametric alternatives, even when the researcher’s main interest is flexible calibration. Figure 1 shows the fitted maps for a fixed illustrative seed, rather than selecting a visually favorable replicate.

The paired comparisons expose variation across designs and classifiers (Figure 2). In the real-data logistic-regression configuration, leaving the scores uncalibrated improves Brier score over isotonic by 0.0013, with a descriptive interval of $[0.0003, 0.0025]$. Calibration therefore has a cost when estimation noise outweighs the correction. An interval crossing zero is not evidence of equivalence, and a positive interval in one configuration does not establish superiority across applications.

Figure 3 puts granularity and Brier score on separate axes for four fixed designs. Rounding the input scores limits the distinctions any deterministic map can retain. Small calibration samples make pooling and flexible fitting different sources of estimation error. Nonmonotone truth falls outside the shape restriction of the monotone methods. The complete configuration-level results, including logarithmic score, AUC, truth error, and runtime, appear in Appendix C; none of these comparisons measures deployment utility.

5 Validation, limits, and practical use

Calibre’s tests check numerical behavior independently of the manuscript. Committed R fixtures compare PAVA, centered isotonic regression, and the nearly isotonic path with reference implementations. Optimizer-based tests check constrained projections. Further tests check score identities, tie handling, fitted-estimator contracts, degenerate inputs, and plotting semantics. Smooth-error fixtures compare with the authors’ reference software. Monte Carlo tests use known generating processes to examine bias and interval coverage in specified settings. Passing these tests establishes agreement on those cases, not universal finite-sample guarantees.

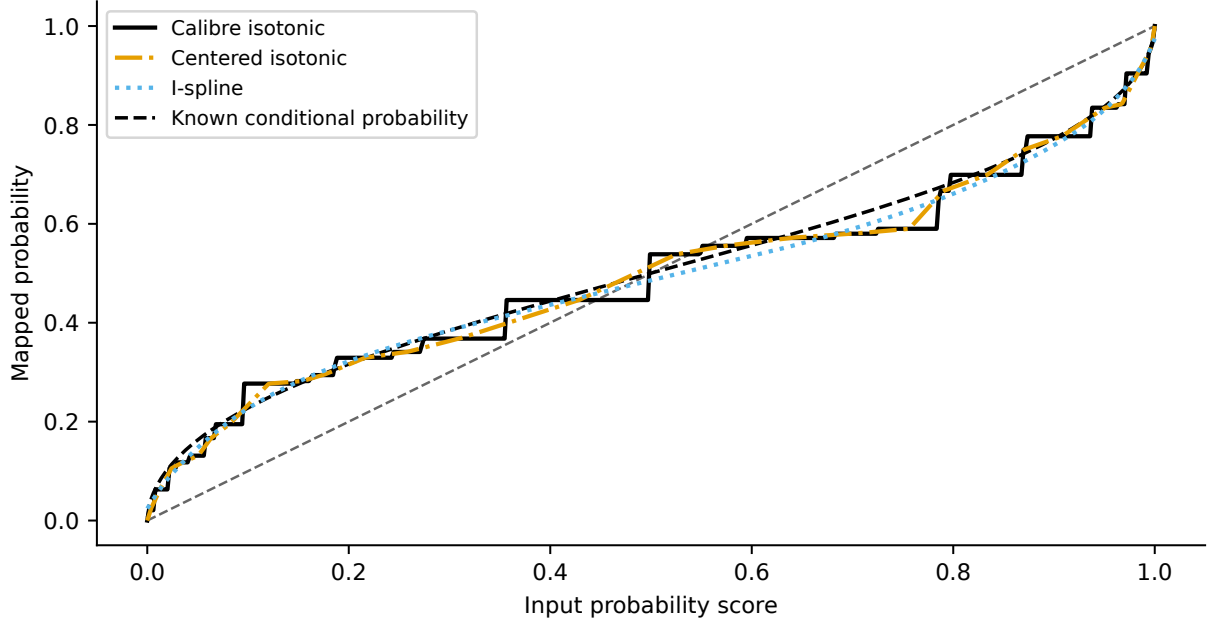


Figure 1: Calibration maps for the overconfidence design, seed zero, fitted on the calibration subset. The dashed curve is the known conditional event probability; the identity line represents no recalibration. Curves show interpolation and pooling behavior, not confidence bands or a separate performance experiment.

Every benchmark cell checks that Calibre’s isotonic wrapper agrees with scikit-learn isotonic predictions to numerical tolerance. The manuscript generator requires the complete offline grid, rejects duplicate records, checks score decompositions, and generates tables and inline numbers from the raw results. A manifest records the input hash and benchmark environment. The paper build uses those results without downloading data or silently rerunning the grid. Appendix D gives the commands.

The benchmark is deliberately limited. 7 synthetic mechanisms and one real dataset cannot establish a general ranking of calibration methods. The nominally matched defaults have different tuning costs, repeated splits are not independent datasets, and score granularity is not a utility measure. A scalar monotone map cannot repair all subgroup errors or recover distinctions absent from its input. The package’s shuffled cross-validation and resampling tools also do not justify inference under temporal or clustered dependence.

For a new application, compare uncalibrated predictions and a simple parametric correction with flexible candidates on a proper score. Use centered isotonic when interpolation between pooled rates is a defensible assumption, and splines when a smooth monotone map is appropriate. Permit decreases only when that departure from ordering is substantively justified and improves validation performance. Examine reliability and ranking alongside the score, and reserve an untouched sample for the final comparison. A decision-utility claim requires explicit decisions, costs, and an evaluation at those operating conditions; none follows from preserving more probability values alone.

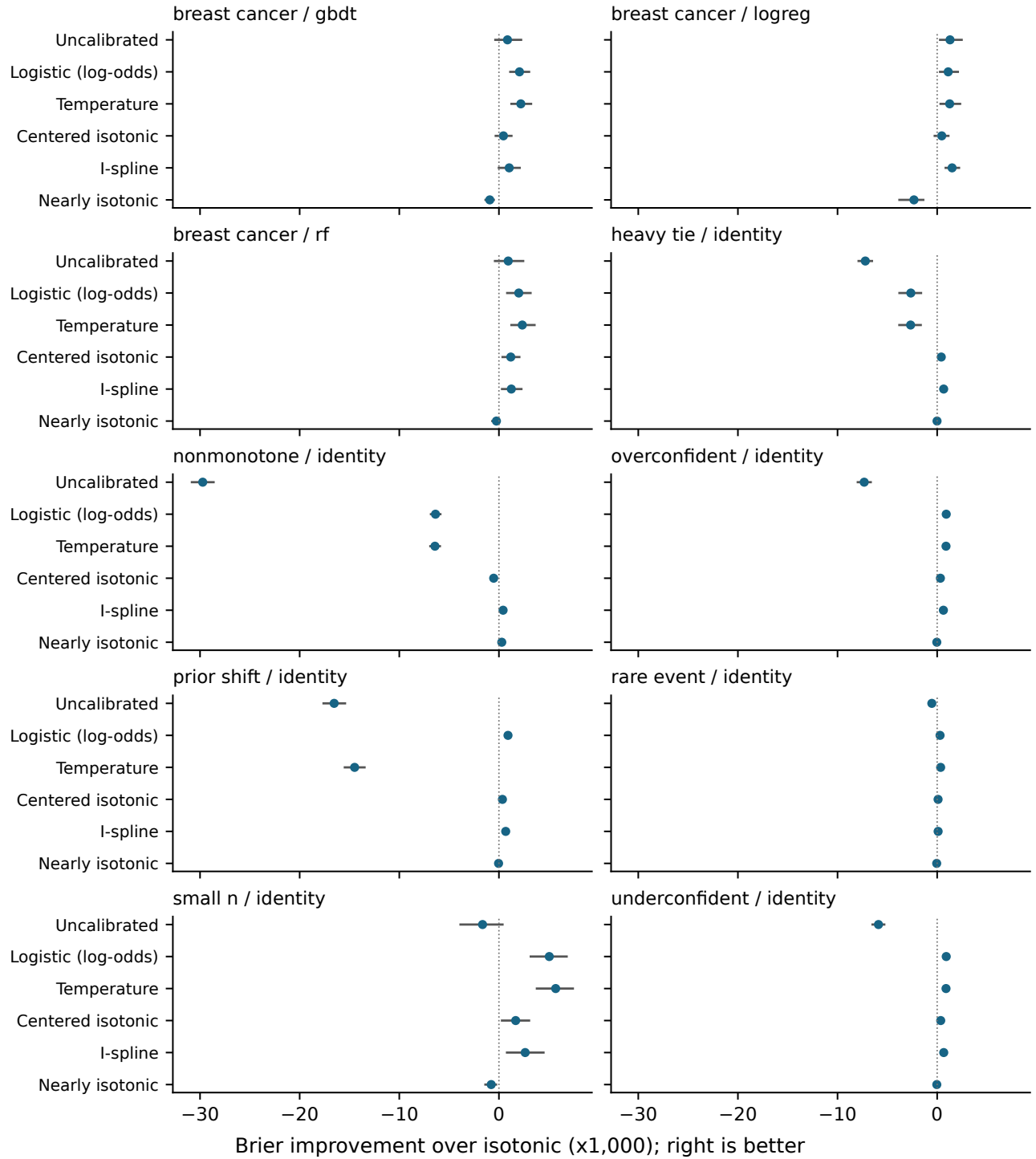


Figure 2: Paired Brier improvements over isotonic for every offline configuration. Dots are mean differences; lines are 95% percentile intervals from 2000 resamples of 30 paired seed differences. All panels share an axis, in thousandths of Brier score. Positive values favor the named method. For breast-cancer data the intervals condition on the same dataset and summarize repeated-split variability. The duplicate isotonic wrapper is omitted.

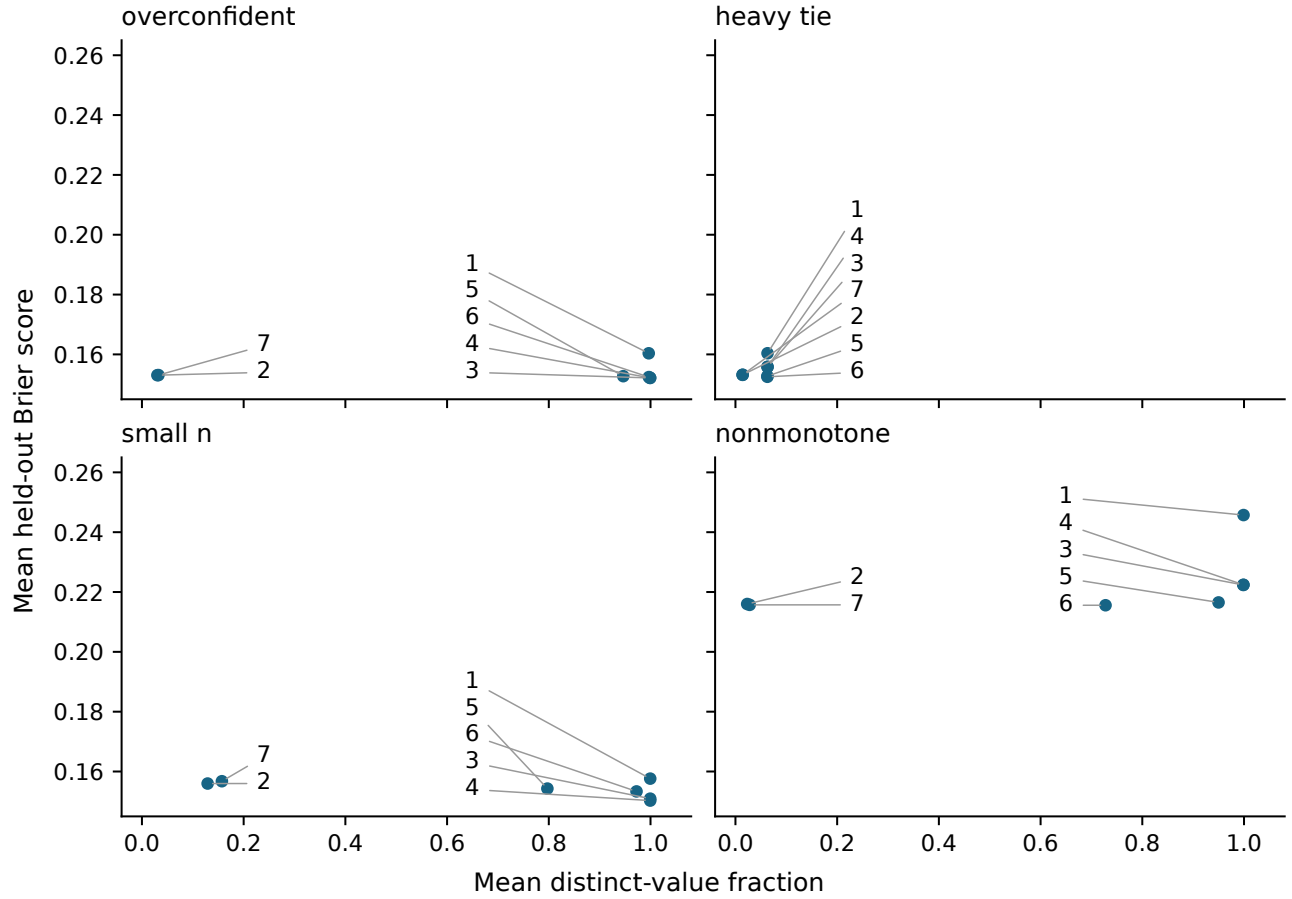


Figure 3: Mean held-out Brier score against the mean fraction of distinct predictions for four fixed illustrative designs, using 30 seeds. Distinct values use six-decimal rounding; the denominator is the test-sample size in each seed. Both axes are shared; thin leader lines connect numbered labels to points. These are descriptive means, with paired score uncertainty reported in Figure 2. Methods are numbered: 1 uncalibrated, 2 isotonic, 3 logistic on log-odds, 4 temperature, 5 centered isotonic, 6 I-spline, 7 nearly isotonic. The duplicate isotonic wrapper is omitted.

A Adjacent increment constraints

For sorted unique score groups with $W_j > 0$, consider

$$\min_{z \in \mathbb{R}^m} \sum_j W_j (\bar{y}_j - z_j)^2 \quad \text{subject to} \quad z_{j+1} - z_j \geq L_j.$$

Set $R_1 = 0$, $R_{j+1} = R_j + L_j$, and $u_j = z_j - R_j$. The constraints become $u_{j+1} \geq u_j$ and the loss is $\sum_j W_j (\bar{y}_j - R_j - u_j)^2$. Thus one weighted isotonic projection gives $z_j = u_j + R_j$. This is the total-order cumulative-shift construction associated with margin-separated isotonic projection (Gunasekar et al., 2016). The substitution supplies the proof directly; it is not evidence that any chosen bound improves calibration.

The projection takes $O(m)$ time once scores are ordered and ties aggregated. Sorting arbitrary observations generally takes $O(n \log n)$. The unbounded feasible set is nonempty for any finite L_j . Adding a probability box can make positive increment requirements infeasible: required cumulative increases can exceed one. Calibre instead solves the unbounded problem and optionally clips its result to $[0, 1]$. Clipping preserves nonpositive lower bounds but can flatten positive increments, so the clipped output is not generally the constrained optimum with a probability box. Bounds on adjacent unique scores also do not establish a derivative bound between scores.

B Experimental cost- and data-informed isotonic calibration

`CDIIIsotonicCalibrator` constructs local bounds from repeated-score event rates and user-supplied thresholds. It is an experimental mapping rule, not a validated procedure for improving economic decisions. Thresholds and input scores must use the same probability-score scale; the kernel does not locate a threshold by inverting a fitted calibration map.

For normalized threshold weights a_k and bandwidth h , the implementation uses

$$e_j = \sum_k a_k \max\{0, 1 - |\tfrac{1}{2}(x_j + x_{j+1}) - t_k|/h\}.$$

Within each unique-score group, Kish effective size is $n_j^{\text{eff}} = (\sum_i w_i)^2 / \sum_i w_i^2$. Adjacent groups define a pooled rate $\tilde{p}_j$, weighted by these effective sizes, and

$$q_j = \sqrt{\tilde{p}_j(1 - \tilde{p}_j) \left(1/n_j^{\text{eff}} + 1/n_{j+1}^{\text{eff}}\right)}.$$

Writing $c = \Phi^{-1}(1 - \alpha/2)$, the bound is

$$L_j = \gamma e_j [\bar{y}_{j+1} - \bar{y}_j - c q_j]_+ - (1 - e_j) c q_j.$$

The cumulative-shift solver then fits a right-continuous step map. Away from the supplied thresholds, the bound can be negative; the result need not be monotone. The clipping limitation in Appendix A applies here too.

Most unique-score groups have size one when scores are continuous. A pooled normal approximation then provides weak evidence, especially near event rates zero and one. Effective sample size makes the calculation invariant to common weight rescaling but does not justify arbitrary sampling or importance weights as binomial replication. Thresholds, bandwidth, and the increment multiplier require application-specific validation. No independent reference implementation or decision-curve experiment is supplied by the current benchmark, so CDI-ISO is excluded from its comparative conclusions.

C Complete benchmark results

Table 3 gives every method/configuration mean. Brier, MCB, and log loss describe accuracy and calibration under their respective scores; AUC measures ranking. Distinct counts describe granularity. Truth MAE exists only in synthetic designs, with the coarsening limitation described in Section 4. Time includes calibrator fitting, internal tuning, and transformation, but excludes base-model fitting and metric evaluation. Runtime depends on the recorded machine and concurrent workload. Neither runtime nor the secondary measures are used to select a headline winner.

Table 3: Complete offline benchmark means over 30 seeds. Counts are rounded to integers; unavailable truth errors are marked $-$. Times below a millisecond are shown as < 0.001 seconds. Paired Brier intervals are in Figure 2.

Method	Brier	MCB	Log loss	AUC	Distinct	Truth MAE	Seconds
<i>breast_cancer / gbd</i>							
Uncalibrated	0.0307	0.0082	0.1431	0.9903	191	$-$	< 0.001
Isotonic	0.0316	0.0058	0.2886	0.9865	12	$-$	< 0.001
Logistic (log-odds)	0.0295	0.0070	0.1129	0.9903	226	$-$	0.001
Temperature	0.0294	0.0069	0.1119	0.9903	227	$-$	< 0.001
Calibre isotonic	0.0316	0.0058	0.2886	0.9865	12	$-$	< 0.001
Centered isotonic	0.0311	0.0081	0.2938	0.9871	78	$-$	< 0.001
I-spline	0.0305	0.0080	0.1309	0.9904	203	$-$	0.426
Nearly isotonic	0.0325	0.0066	0.4036	0.9824	20	$-$	0.024
<i>breast_cancer / logreg</i>							
Uncalibrated	0.0208	0.0058	0.0817	0.9941	196	$-$	< 0.001
Isotonic	0.0221	0.0035	0.2965	0.9892	9	$-$	< 0.001
Logistic (log-odds)	0.0209	0.0060	0.0900	0.9941	167	$-$	< 0.001
Temperature	0.0208	0.0059	0.0896	0.9941	171	$-$	< 0.001
Calibre isotonic	0.0221	0.0035	0.2965	0.9892	9	$-$	< 0.001
Centered isotonic	0.0216	0.0050	0.2949	0.9898	56	$-$	< 0.001
I-spline	0.0206	0.0056	0.1006	0.9941	167	$-$	0.312
Nearly isotonic	0.0244	0.0044	0.4592	0.9840	13	$-$	0.013
<i>breast_cancer / rf</i>							
Uncalibrated	0.0346	0.0083	0.1587	0.9882	80	$-$	< 0.001
Isotonic	0.0355	0.0058	0.3043	0.9843	10	$-$	< 0.001
Logistic (log-odds)	0.0335	0.0073	0.1592	0.9882	80	$-$	< 0.001
Temperature	0.0332	0.0069	0.1576	0.9882	80	$-$	< 0.001
Calibre isotonic	0.0355	0.0058	0.3043	0.9843	10	$-$	< 0.001
Centered isotonic	0.0343	0.0077	0.3009	0.9851	48	$-$	< 0.001
I-spline	0.0343	0.0077	0.1424	0.9878	76	$-$	0.331
Nearly isotonic	0.0358	0.0062	0.3050	0.9841	11	$-$	0.022
<i>heavy_tie / identity</i>							
Uncalibrated	0.1604	0.0111	0.6380	0.8603	101	0.0810	< 0.001
Isotonic	0.1532	0.0022	0.4685	0.8590	22	0.0265	< 0.001
Logistic (log-odds)	0.1558	0.0066	0.5142	0.8603	101	0.0524	0.002
Temperature	0.1558	0.0066	0.5142	0.8603	101	0.0523	< 0.001
Calibre isotonic	0.1532	0.0022	0.4685	0.8590	22	0.0265	< 0.001
Centered isotonic	0.1528	0.0035	0.4673	0.8603	101	0.0216	< 0.001
I-spline	0.1525	0.0033	0.4665	0.8603	100	0.0189	0.905
Nearly isotonic	0.1532	0.0023	0.4685	0.8588	23	0.0267	0.074

Table 3 continued

Method	Brier	MCB	Log loss	AUC	Distinct	Truth MAE	Seconds
<i>nonmonotone / identity</i>							
Uncalibrated	0.2457	0.0339	0.6921	0.6754	1598	0.1589	< 0.001
Isotonic	0.2160	0.0022	0.6347	0.6820	37	0.0354	< 0.001
Logistic (log-odds)	0.2224	0.0106	0.6304	0.6754	1598	0.0792	0.002
Temperature	0.2224	0.0106	0.6306	0.6754	1598	0.0792	< 0.001
Calibre isotonic	0.2160	0.0022	0.6347	0.6820	37	0.0354	0.001
Centered isotonic	0.2165	0.0047	0.6358	0.6754	1520	0.0405	0.004
I-spline	0.2156	0.0038	0.6145	0.6796	1164	0.0304	1.087
Nearly isotonic	0.2157	0.0025	0.6341	0.6905	45	0.0310	0.536
<i>overconfident / identity</i>							
Uncalibrated	0.1604	0.0121	0.5187	0.8606	1594	0.0808	< 0.001
Isotonic	0.1530	0.0025	0.4891	0.8593	49	0.0255	< 0.001
Logistic (log-odds)	0.1521	0.0039	0.4643	0.8606	1599	0.0064	0.002
Temperature	0.1522	0.0039	0.4643	0.8606	1599	0.0040	< 0.001
Calibre isotonic	0.1530	0.0025	0.4891	0.8593	49	0.0255	< 0.001
Centered isotonic	0.1527	0.0044	0.4885	0.8606	1514	0.0205	0.003
I-spline	0.1524	0.0042	0.4660	0.8606	1595	0.0175	0.987
Nearly isotonic	0.1531	0.0025	0.4892	0.8590	52	0.0259	0.408
<i>prior_shift / identity</i>							
Uncalibrated	0.1696	0.0213	0.5104	0.8606	1599	0.1190	< 0.001
Isotonic	0.1530	0.0025	0.4891	0.8593	49	0.0255	< 0.001
Logistic (log-odds)	0.1521	0.0039	0.4643	0.8606	1599	0.0064	0.002
Temperature	0.1675	0.0193	0.5025	0.8606	1599	0.1077	< 0.001
Calibre isotonic	0.1530	0.0025	0.4891	0.8593	49	0.0255	< 0.001
Centered isotonic	0.1527	0.0044	0.4882	0.8606	1514	0.0199	0.004
I-spline	0.1524	0.0041	0.4662	0.8606	1596	0.0166	1.385
Nearly isotonic	0.1531	0.0025	0.4892	0.8590	52	0.0259	0.483
<i>rare_event / identity</i>							
Uncalibrated	0.0245	0.0019	0.1220	0.8390	1144	0.0185	< 0.001
Isotonic	0.0239	0.0009	0.1236	0.8317	23	0.0076	< 0.001
Logistic (log-odds)	0.0236	0.0011	0.0991	0.8390	1558	0.0027	0.002
Temperature	0.0236	0.0011	0.0988	0.8390	1560	0.0017	< 0.001
Calibre isotonic	0.0239	0.0009	0.1236	0.8317	23	0.0076	< 0.001
Centered isotonic	0.0238	0.0012	0.1231	0.8375	1189	0.0061	0.003
I-spline	0.0238	0.0013	0.1009	0.8390	1407	0.0067	0.960
Nearly isotonic	0.0240	0.0009	0.1240	0.8265	30	0.0083	0.495
<i>small_n / identity</i>							
Uncalibrated	0.1576	0.0263	0.4996	0.8644	120	0.0810	< 0.001
Isotonic	0.1560	0.0144	0.7758	0.8573	16	0.0640	< 0.001
Logistic (log-odds)	0.1509	0.0196	0.4597	0.8644	120	0.0283	0.001
Temperature	0.1503	0.0190	0.4578	0.8644	120	0.0187	< 0.001
Calibre isotonic	0.1560	0.0144	0.7758	0.8573	16	0.0640	< 0.001
Centered isotonic	0.1543	0.0225	0.7752	0.8635	96	0.0579	< 0.001
I-spline	0.1533	0.0220	0.5036	0.8641	117	0.0430	0.465
Nearly isotonic	0.1568	0.0150	0.7786	0.8537	19	0.0675	0.035

Table 3 continued

Method	Brier	MCB	Log loss	AUC	Distinct	Truth MAE	Seconds
<i>underconfident / identity</i>							
Uncalibrated	0.1589	0.0107	0.4878	0.8606	1599	0.0772	< 0.001
Isotonic	0.1530	0.0025	0.4891	0.8593	49	0.0255	< 0.001
Logistic (log-odds)	0.1521	0.0039	0.4643	0.8606	1599	0.0064	0.002
Temperature	0.1522	0.0039	0.4643	0.8606	1599	0.0040	< 0.001
Calibre isotonic	0.1530	0.0025	0.4891	0.8593	49	0.0255	< 0.001
Centered isotonic	0.1527	0.0044	0.4881	0.8606	1514	0.0198	0.004
I-spline	0.1524	0.0041	0.4663	0.8606	1595	0.0166	1.407
Nearly isotonic	0.1531	0.0025	0.4892	0.8590	52	0.0259	0.483

D Reproduction and software availability

The package and benchmark are available at <https://github.com/finite-sample/calibre>. Installation and the executable tutorials are documented at <https://finite-sample.github.io/calibre/>. The software uses the MIT license. The following commands install the locked environment, rerun the offline grid, rebuild its shared summaries, and compile this paper:

```
uv sync --locked --all-groups
uv run python -m benchmarks.run --n-jobs 4
uv run python -m benchmarks.aggregate
make paper
```

`make paper` generates exhibits from the raw benchmark and uses `latexmk`; it requires a LaTeX installation. Its output is `ms/build/calibre.pdf`. `make paper-check` executes the example and the manuscript artifact tests. The benchmark environment stores versions, revision, and source hashes, including the lockfile. The generated manifest links that environment to the raw input hash. Changes to defaults require rerunning the grid before making new empirical claims.

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